

DEUTSCHE BANK INDIA

Retail Banking — Digital Payments

UAT SCENARIOS

UPI Enablement Platform

User Acceptance Test Scripts Across All 10 Modules and Non-Functional Requirements

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Document Control

1.1 Version History

Version	Date	Author	Description
0.1	10-Jul-2026	Business Analyst	Initial test case draft derived from FRS and BRD Section 17
1.0	date TBD	Business Analyst / QA Lead	Baseline for UAT execution

1.2 Relationship to Other Deliverables

These scenarios execute against the functional requirements (FR-xx), business rules (BR-xx), and error codes (ERR-xx) defined in the FRS, and validate the use cases in the Use Case Catalogue. The BRD (Section 17) defines the overall UAT approach and entry/exit criteria, restated here for execution reference; this document provides the executable test scripts themselves.

Table of Contents

Document Control.....2

 1.1 Version History.....2

 1.2 Relationship to Other Deliverables2

Table of Contents3

2. UAT Approach Summary.....4

 2.1 Entry Criteria4

 2.2 Exit Criteria.....4

 2.3 Test Case Format.....4

3. Module 1 — Customer Registration5

4. Module 2 — Bank Account Linking.....6

5. Module 3 — UPI PIN Management7

6. Module 4 — Send Money8

7. Module 5 — Collect Requests.....9

8. Module 6 — QR Payments.....10

9. Module 7 — Balance Check11

10. Module 8 — Transaction History.....12

11. Module 9 — Notifications.....13

12. Module 10 — Dispute Management14

13. Non-Functional Test Scenarios15

14. Traceability and Sign-Off16

 14.1 Execution Sign-Off.....16

2. UAT Approach Summary

This restates the UAT approach defined in BRD Section 17 for execution reference. Full scope, roles, and sign-off responsibilities remain governed by the BRD.

2.1 Entry Criteria

- All Must Have functional requirements are code-complete and have passed System Integration Testing (SIT).
- NPCI UAT (sandbox) environment access is confirmed and connectivity validated.
- Representative test data (customer profiles, accounts, devices) is provisioned in the UAT environment.
- No open Critical or High severity defects remain from SIT.

2.2 Exit Criteria

- 100% of Must Have test cases executed with no open Critical/High severity defects.
- $\geq 95\%$ of Should Have test cases executed and passed.
- Business Sponsor, Compliance, and Risk sign-off obtained per the BRD RACI matrix.
- NPCI production certification confirmed as a precondition to go-live, independent of internal UAT sign-off.

2.3 Test Case Format

Each test case specifies a unique ID (prefixed by module), the scenario under test with its related requirement ID in parentheses, step-by-step test actions, the expected result, MoSCoW priority, and a blank Pass/Fail column for execution recording. Preconditions are included as the first test step where not otherwise implied by the scenario.

3. Module 1 — Customer Registration

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-REG-01	Successful UPI registration (FR-REG-01 to 09)	1. Log in to mobile app. 2. Tap Register for UPI. 3. Verify OTP. 4. Accept consent. 5. Select eligible account(s) and default. 6. Confirm.	UPI profile created; VPA assigned; success screen shown; confirmation notification received.	Must	
TC-REG-02	Registration blocked on mobile mismatch (ERR-REG-004)	1. Initiate registration with a mobile number not matching CBS/CIF records. 2. Attempt verification.	Registration halted with remediation message; no UPI profile created.	Must	
TC-REG-03	OTP expiry handling (ERR-REG-001)	1. Initiate registration. 2. Wait beyond 300 seconds. 3. Enter the expired OTP.	System rejects the OTP with an expiry message and offers resend.	Must	
TC-REG-04	Consent decline halts registration (FR-REG-06)	1. Initiate registration through mobile verification. 2. On the consent screen, select Decline.	Registration halted; no Consent record with Accept status created; customer returned to UPI landing.	Must	
TC-REG-05	No eligible accounts found (ERR-REG-005)	1. Initiate registration for a customer with no active savings/current account.	System displays 'no eligible account' message and terminates the registration flow.	Must	
TC-REG-06	Custom VPA handle selection (FR-REG-10)	1. During registration, select 'Customise UPI ID'. 2. Enter an available handle. 3. Confirm.	Handle is reserved and assigned as the profile's VPA.	Should	

4. Module 2 — Bank Account Linking

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-ACC-01	Link additional eligible account (FR-ACC-03)	1. Open Manage Linked Accounts. 2. Tap Add Account. 3. Select an eligible unlinked account. 4. Confirm.	Account appears in linked accounts list with link_status Active.	Must	
TC-ACC-02	Blocked: remove default without reassignment (ERR-ACC-001)	1. With 2+ linked accounts, attempt to remove the current default account.	System blocks the action and prompts to set a new default first.	Must	
TC-ACC-03	Change default account (FR-ACC-05)	1. With 2+ linked accounts, select 'Set as Default' on a non-default account.	New account becomes default; prior default account's is_default flag cleared.	Must	
TC-ACC-04	Blocked: ineligible account type (ERR-ACC-002)	1. Attempt to link a loan or credit card account.	System blocks linking with an ineligible-type message.	Must	
TC-ACC-05	Automatic delinking on closure (FR-ACC-07)	1. Simulate a linked account status change to Closed in CBS. 2. Trigger reconciliation batch/event.	Account automatically delinked; customer notified; if it was default, prompted to select a new default.	Must	

5. Module 3 — UPI PIN Management

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-PIN-01	Successful PIN set via debit card (FR-PIN-01 to 04)	1. Initiate Set PIN. 2. Enter valid debit card last 6 digits + expiry. 3. Verify OTP. 4. Enter and confirm new PIN.	PIN set successfully; confirmation notification sent; no plaintext PIN stored anywhere.	Must	
TC-PIN-02	PIN set via Aadhaar-OTP (FR-PIN-01)	1. Initiate Set PIN. 2. Select Aadhaar-OTP path. 3. Verify Aadhaar-OTP. 4. Set new PIN.	PIN set successfully without requiring debit card details.	Should	
TC-PIN-03	PIN reset without old PIN (FR-PIN-05)	1. Select Forgot PIN. 2. Validate via debit card/OTP. 3. Set new PIN.	PIN reset successfully; old PIN not required at any step.	Must	
TC-PIN-04	PIN change requires correct old PIN (FR-PIN-06)	1. Select Change PIN. 2. Enter incorrect current PIN.	System rejects the change and does not proceed to new PIN entry.	Must	
TC-PIN-05	Lockout after 3 incorrect attempts (ERR-PIN-005)	1. Attempt any PIN-authenticated action. 2. Enter incorrect PIN 3 consecutive times.	Profile locked for 24 hours; security alert notification sent; further PIN actions blocked.	Must	
TC-PIN-06	Auto-unlock after 24 hours (FR-PIN-07)	1. With a locked profile, advance system time (or wait) past the 24-hour window. 2. Attempt a PIN-authenticated action.	Profile is usable again without requiring an explicit reset.	Should	

6. Module 4 — Send Money

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-TXN-01	Successful payment via UPI ID (FR-TXN-02)	1. Select Send Money > UPI ID. 2. Enter valid VPA. 3. Confirm recipient name. 4. Enter amount. 5. Authorise with PIN.	Payment completes; status Success with NPCI RRN shown; notification sent; balance updated.	Must	
TC-TXN-02	Successful payment via mobile number (FR-TXN-01)	1. Select Send Money > Mobile Number. 2. Enter a UPI-registered number. 3. Confirm resolved VPA. 4. Enter amount. 5. Authorise with PIN.	Payment completes successfully as above.	Must	
TC-TXN-03	Successful payment via Bank Account + IFSC (FR-TXN-04)	1. Select Send Money > Bank Account + IFSC. 2. Enter valid account number and IFSC. 3. Enter amount. 4. Authorise with PIN.	Payment completes successfully via account-based UPI transfer.	Must	
TC-TXN-04	Self-transfer between own accounts (FR-TXN-05)	1. Select Send Money > Self Transfer. 2. Select source and destination linked accounts. 3. Enter amount. 4. Authorise with PIN.	Funds move between the customer's own accounts; payee-specific limits not applied (BR-TXN-07).	Should	
TC-TXN-05	Blocked: insufficient balance (ERR-TXN-007)	1. Initiate a payment exceeding available balance. 2. Authorise with PIN.	Transaction is blocked before submission to NPCI with an insufficient-balance message.	Must	
TC-TXN-06	Blocked: exceeds per-transaction limit (ERR-TXN-005)	1. Initiate a payment above ₹1,00,000 (or configured limit). 2. Attempt to authorise.	Transaction is blocked with a limit-exceeded message before NPCI submission.	Must	
TC-TXN-07	Declined by fraud engine (ERR-TXN-008)	1. Initiate a payment matching a configured high-risk fraud rule in the test environment. 2. Authorise with PIN.	Transaction is declined post risk-scoring, before NPCI submission, with a generic security message.	Must	
TC-TXN-08	NPCI timeout → Pending → auto-reversal (FR-TXN-14, BR-TXN-08)	1. Initiate a payment against a simulated NPCI timeout in the test environment. 2. Observe status. 3. Allow the Deferred Status Check job to run.	Transaction shows Pending initially; if debit-without-credit is confirmed, auto-reversal occurs and customer is notified.	Must	

7. Module 5 — Collect Requests

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-COL-01	Create and approve collect request (FR-COL-01, FR-COL-03)	1. Requester creates a collect request. 2. Payer receives notification. 3. Payer approves with PIN.	Payment processes as a standard transaction; Collect_Request status set to Approved.	Must	
TC-COL-02	Reject collect request with reason (FR-COL-04)	1. Payer opens a Pending request. 2. Selects Reject with an optional reason.	Request status set to Rejected; requester notified with the reason if provided.	Must	
TC-COL-03	Auto-expire after 24 hours (FR-COL-05, BR-COL-01)	1. Create a collect request. 2. Do not action it. 3. Allow 24 hours to elapse (or advance test clock). 4. Run expiry job.	Request status automatically set to Expired; requester notified.	Must	
TC-COL-04	Reminder limit enforcement (FR-COL-06, BR-COL-04)	1. Create a collect request. 2. Send reminders up to 3 times. 3. Attempt a 4th reminder.	First 3 reminders send successfully; 4th is blocked/disabled with an explanatory message.	Should	
TC-COL-05	Blocked: request already expired (ERR-COL-001)	1. Attempt to approve a collect request after its expiry.	System blocks the action with an expired-request message.	Must	

8. Module 6 — QR Payments

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-QR-01	Generate static QR and receive payment (FR-QR-01)	1. Open My QR Code. 2. Select Static. 3. Share/display the QR. 4. Have another UPI app scan and pay a chosen amount.	Payment is credited to the linked default account; QR does not embed a fixed amount.	Must	
TC-QR-02	Generate dynamic QR with fixed amount (FR-QR-02)	1. Open My QR Code. 2. Select Dynamic. 3. Enter an amount. 4. Generate and display QR.	QR embeds the fixed amount, non-editable by the payer; expires per the configured validity window.	Must	
TC-QR-03	Scan & pay to verified merchant (FR-QR-03, FR-QR-04)	1. Select Scan & Pay. 2. Scan a valid, NPCI-registered merchant QR. 3. Confirm amount. 4. Authorise with PIN.	Merchant is validated against the NPCI registry; payment completes successfully.	Must	
TC-QR-04	Blocked: unverified/blacklisted merchant (ERR-QR-002)	1. Scan a QR resolving to a merchant VPA flagged as blacklisted in the test environment.	Payment is blocked before amount entry, with a clear reason shown.	Must	
TC-QR-05	Blocked: expired dynamic QR (ERR-QR-003)	1. Generate a dynamic QR. 2. Wait beyond its validity window. 3. Attempt to scan and pay.	System rejects the expired QR with a clear message.	Should	

9. Module 7 — Balance Check

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-BAL-01	Successful balance check with PIN (FR-BAL-01)	1. Select Check Balance. 2. Choose an account. 3. Authorise with PIN.	Available and current balance displayed accurately for the selected account.	Must	
TC-BAL-02	Daily cap enforcement (ERR-BAL-001, BR-BAL-02)	1. Perform 15 balance checks in one day. 2. Attempt a 16th.	16th attempt is blocked with a daily-limit message; no enquiry submitted to NPCI/CBS.	Should	
TC-BAL-03	No cached balance without fresh auth (BR-BAL-01)	1. Check balance successfully. 2. Close and reopen the balance screen without re-entering PIN.	No balance value is displayed until PIN is re-entered; no client-side cache is shown.	Must	

10. Module 8 — Transaction History

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-HIS-01	View default 6-month history (FR-HIS-01)	1. Open Transaction History with no filters applied.	Most recent 6 months of transactions displayed, sorted descending by date.	Must	
TC-HIS-02	Combined filter application (FR-HIS-02)	1. Apply a date range, status = Success, and a beneficiary name filter together.	Only transactions matching all three criteria simultaneously are returned.	Must	
TC-HIS-03	Export PDF statement (FR-HIS-03)	1. Select a date range. 2. Tap Export PDF.	PDF statement generated matching the active filters, downloadable/shareable.	Should	
TC-HIS-04	Transaction detail drill-down (FR-HIS-04)	1. Tap a transaction from the list.	Full detail shown including internal transaction ID, NPCI RRN, timestamps, and status.	Must	

11. Module 9 — Notifications

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-NOT-01	Payment success notification timing (FR-NOT-01, BR-NOT-01)	1. Complete a successful payment. 2. Measure time to notification receipt.	Push/SMS notification received within 5 seconds of transaction status determination.	Must	
TC-NOT-02	Security alert cannot be opted out (BR-NOT-02)	1. Disable all optional notification toggles available. 2. Trigger a new-device security event.	Security alert is still delivered regardless of preference settings.	Must	
TC-NOT-03	Marketing opt-out honoured (FR-NOT-07)	1. Disable marketing/cross-sell notifications. 2. Trigger a cross-sell offer event.	No marketing notification is sent; transaction notifications remain unaffected.	Should	
TC-NOT-04	Push failure falls back to SMS for security event (FRS 11.1)	1. Simulate push delivery failure in test environment for a PIN lockout event.	System falls back to SMS delivery for this security-critical event.	Should	
TC-NOT-05	Collect request notification delivered (FR-NOT-03)	1. Create a collect request against a test payer.	Payer receives a real-time notification of the new request.	Must	

12. Module 10 — Dispute Management

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-DIS-01	Raise dispute within 180 days (FR-DIS-01, FR-DIS-02)	1. Select a transaction less than 180 days old. 2. Raise Dispute. 3. Select category, enter description. 4. Submit.	Dispute created with Submitted status; internal reference number generated immediately.	Must	
TC-DIS-02	Blocked: outside 180-day window (ERR-DIS-001)	1. Attempt to raise a dispute on a transaction older than 180 days (e.g., via deep link).	System blocks the action with an outside-window message.	Must	
TC-DIS-03	Blocked: duplicate dispute already open (ERR-DIS-002)	1. Raise a dispute on a transaction. 2. Attempt to raise a second dispute on the same transaction while the first remains open.	Second attempt is blocked, directing the customer to existing Dispute Tracking.	Should	
TC-DIS-04	NPCI reference captured on escalation (FR-DIS-04)	1. Raise a dispute in a category requiring NPCI escalation. 2. Allow escalation to process.	NPCI complaint/reference number is captured and displayed in Dispute Tracking.	Must	
TC-DIS-05	Refund processed and status displayed (FR-DIS-06)	1. Have Operations uphold a test dispute and initiate a refund.	Refund status and credit date are visible to the customer in Dispute Tracking.	Must	
TC-DIS-06	Auto-escalation on TAT breach (FR-DIS-07, BR-DIS-03)	1. Leave a test dispute unresolved beyond its category TAT threshold. 2. Allow the escalation job to run.	Dispute is automatically flagged for supervisor escalation.	Should	

13. Non-Functional Test Scenarios

TC ID	Scenario	Test Steps	Expected Result	Pri.	P/F
TC-NFR-01	Average transaction latency (NFR-PERF-01)	1. Execute 100 representative Send Money transactions under normal load. 2. Measure end-to-end time.	Average completion time $\leq$ 8 seconds; no individual transaction exceeds the 30-second NPCI SLA ceiling.	Must	
TC-NFR-02	Peak load throughput (NFR-PERF-02)	1. Execute a load test simulating 500 transactions per second sustained for 10 minutes.	Platform processes the load without error-rate degradation beyond agreed thresholds.	Must	
TC-NFR-03	PIN never stored/logged in plaintext (NFR-SEC-02)	1. Perform a PIN set/change/reset. 2. Inspect application logs and database records for the PIN value.	No plaintext or reversible PIN value found in any log, database table, or audit record.	Must	
TC-NFR-04	Penetration test — no critical/high findings (NFR-SEC-05)	1. Execute the pre-go-live penetration test per the agreed scope.	Zero critical or high-severity findings remain open at go-live.	Must	
TC-NFR-05	Failover — PSP switch / API gateway (NFR-AVL-02)	1. Simulate failure of the primary PSP switch/API gateway node during active transaction load.	Traffic fails over automatically with no customer-visible outage beyond the defined threshold.	Should	
TC-NFR-06	Disaster recovery — RTO/RPO (NFR-AVL-03)	1. Execute a planned DR failover test. 2. Measure recovery time and data loss window.	Recovery Time Objective $\leq$ 4 hours; Recovery Point Objective $\leq$ 15 minutes.	Should	
TC-NFR-07	Core flow screen-count (NFR-USE-02)	1. Execute Send Money and Scan & Pay flows end-to-end, counting distinct screens (excluding PIN entry).	Each flow completes in 4 screens or fewer.	Should	
TC-NFR-08	Data residency — India-only hosting (NFR-CMP-03)	1. Review infrastructure/hosting configuration for all customer payment data stores.	All customer PII and transaction data confirmed hosted exclusively within India.	Must	

14. Traceability and Sign-Off

This document contains 61 test cases: 53 functional (across the 10 modules) and 8 non-functional. Every test case traces to a Functional Requirement (FR-xx), Business Rule (BR-xx), or Non-Functional Requirement (NFR-xx) in the FRS, and every module's test cases collectively exercise the corresponding Use Cases in the Use Case Catalogue. Full ID-level cross-referencing is maintained in the Requirement Traceability Matrix (RTM).

14.1 Execution Sign-Off

Role	Responsibility	Sign-Off
QA Lead	Confirms all Must Have cases executed with no open Critical/High defects	Pending
Business Sponsor	Confirms business acceptance of test results	Pending
Compliance	Confirms regulatory-relevant scenarios (TC-DIS, TC-NFR-03, TC-NFR-08) passed	Pending
Risk	Confirms fraud/limit scenarios (TC-TXN-05 to 07) passed	Pending